

# THE ORDER-CONVEX CLONE ON THE BOOLEAN CUBE: UNBOUNDED WIDTH, NP-HARDNESS, AND MEDIAN-CLOSURE AS THE EXACT BOUNDARY

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**ABSTRACT.** The realizability of a cell along a monotone generator of the cube category is a monotone-selection constraint problem whose variable domains are the fibres of the generator. When those fibres are closed under the median the problem carries a majority polymorphism and hence has strict width two; the coskeletal dimension of the corresponding cubical atom is then bounded by the arity of the median rather than by the ambient dimension. We show that this bound has an exact algebraic boundary. Let  $L_m$  be the relational structure on  $\{0, 1\}^m$  whose relations are the order and all order-convex subsets. We prove that for every  $m \geq 3$  the clone  $\text{Pol}(L_m)$  contains *no* ternary weak near-unanimity operation; consequently  $L_m$  does not have bounded width. The proof reduces to the single Boolean cube  $\{0, 1\}^3$  by an order-convex restriction, where the absence is established by hand from a twelve-cell contradiction (a minimal core located by machine). The same restriction, applied to the 4-ary Siggers operation, shows the clone is not even Taylor, so the constraint problem is in fact NP-complete for every  $m \geq 3$  (and in P for  $m \leq 2$ ). Thus median-closure of the fibres is precisely the boundary—between bounded width and unbounded width, and between polynomial-time and NP-hard realizability—and a cubical atom with a non-median-closed generator is unboundedly coskeletal in every dimension.

## 1. INTRODUCTION

The Dedekind cube category  $\square$  is the Lawvere theory of bounded distributive lattices: its morphisms  $[k] \rightarrow [m]$  are the monotone maps  $\{0, 1\}^k \rightarrow \{0, 1\}^m$ . A *cell* of a presheaf is realizable along a monotone generator  $z: \{0, 1\}^m \rightarrow \{0, 1\}^n$  exactly when it factors through  $z$ , and this factorization problem is, vertexwise, a *monotone-selection constraint satisfaction problem*: for a cell  $a: \{0, 1\}^k \rightarrow \{0, 1\}^n$  one seeks a monotone  $g: \{0, 1\}^k \rightarrow \{0, 1\}^m$  with  $z \circ g = a$ , i.e. one variable  $g(v)$  per vertex  $v$  of  $\{0, 1\}^k$ , with domain the fibre  $z^{-1}(a(v))$  and a binary constraint  $g(v) \leq g(v')$  along each edge  $v \leq v'$ . The companion work [1] shows that when every fibre  $z^{-1}(q)$  is closed under the ternary median  $\text{med}(x, y, z) = (x \wedge y) \vee (y \wedge z) \vee (z \wedge x)$ , this constraint problem carries the median as a *majority polymorphism*, so by the Baker–Pixley theorem [2] it has *strict width two*: it is solvable if and only if it is (2, 3)-consistent, and the solution set is determined by its two-variable projections. The associated cubical atom is then determined by its cells of bounded level—its coskeletal dimension is governed by the arity 3 of the median, not by the ambient dimension—so that such atoms are finite in number.

The hypothesis that the fibres be median-closed is essential. A fibre of a monotone map is always *order-convex* (if  $a \leq c \leq b$  and  $z(a) = z(b) = q$  then  $z(c) = q$ ), but need not be median-closed: the antichain  $\{001, 010, 100\} \subseteq \{0, 1\}^3$  has median 000, which lies below all three of its members (Figure 1), so an order-convex set containing the antichain but not 000 is not median-closed. This note settles what happens for general order-convex fibres. The relevant object is the relational structure carrying *all* order-convex relations, and the question is whether it has bounded width in the sense of Feder and Vardi. We prove it does not, in every dimension  $m \geq 3$ , and that the median-closed case is the exact boundary.

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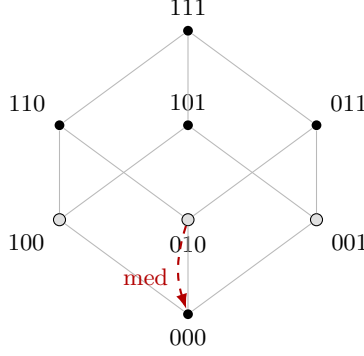


FIGURE 1. The cube  $\{0,1\}^3$ . The three atoms 100, 010, 001 (shaded) form an antichain whose convex hull is themselves; but their median is the bottom 000, which escapes the hull. A ternary majority operation compatible with all order-convex sets is thereby obstructed, and the theorem shows no weak near-unanimity operation avoids the obstruction.

## 2. THE ORDER-CONVEX STRUCTURE AND ITS CLONE

**Definition 2.1.** For  $m \geq 1$  let  $\{0,1\}^m$  carry the componentwise partial order. A set  $C \subseteq \{0,1\}^m$  is *order-convex* if  $a, b \in C$  and  $a \leq c \leq b$  imply  $c \in C$ . Let  $L_m$  be the relational structure on the domain  $\{0,1\}^m$  whose relations are the order  $\leq = \{(x, y) : x \leq y\}$  and every order-convex subset  $C \subseteq \{0,1\}^m$  regarded as a unary relation.

An operation  $f: (\{0,1\}^m)^k \rightarrow \{0,1\}^m$  is a *polymorphism* of  $L_m$  if it preserves every relation; we write  $\text{Pol}(L_m)$  for the clone of all polymorphisms. Preservation unwinds as follows.

**Lemma 2.2.**  $f \in \text{Pol}(L_m)$  if and only if  $f$  is

- (1) monotone:  $x_i \leq y_i$  for all  $i$  implies  $f(x_1, \dots, x_k) \leq f(y_1, \dots, y_k)$ ;
- (2) idempotent:  $f(x, \dots, x) = x$ ; and
- (3) order-convex-preserving:  $f(x_1, \dots, x_k) \in \text{conv}\{x_1, \dots, x_k\}$  for all inputs, where  $\text{conv } X = \{c : \exists a, b \in X, a \leq c \leq b\}$  is the order-convex hull.

*Proof.* Preserving the binary relation  $\leq$  is exactly monotonicity. Each singleton  $\{a\} = [a, a]$  is order-convex, so preserving all unary order-convex relations forces  $f(a, \dots, a) = a$ , idempotence. For the convex-hull condition:  $\text{conv}\{x_1, \dots, x_k\}$  is itself order-convex and contains each  $x_i$ , so if  $f$  preserves it then  $f(x_1, \dots, x_k) \in \text{conv}\{x_1, \dots, x_k\}$ . Conversely, if  $f$  always lands in the convex hull of its arguments and  $C$  is any order-convex set with all  $x_i \in C$ , then  $\text{conv}\{x_1, \dots, x_k\} \subseteq C$ , whence  $f(x_1, \dots, x_k) \in C$ ; thus  $f$  preserves  $C$ .  $\square$

**Definition 2.3.** A  $k$ -ary operation  $w$  ( $k \geq 2$ ) is a *weak near-unanimity* (WNU) operation if it is idempotent and

$$w(y, x, x, \dots, x) = w(x, y, x, \dots, x) = \dots = w(x, x, \dots, x, y) \quad \text{for all } x, y.$$

A *majority* operation is a ternary WNU with  $w(x, x, y) = w(x, y, x) = w(y, x, x) = x$ .

We use the following form of the algebraic characterization of bounded width, due to Barto–Kozik and Kozik–Krokhin–Valeriote–Willard.

**Theorem 2.4** ([3, 4]). *A finite idempotent relational structure has bounded width if and only if it admits weak near-unanimity polymorphisms of all arities  $n \geq 3$ ; equivalently, ternary and 4-ary WNU polymorphisms  $v, w$  with  $v(y, x, x) = w(y, x, x, x)$ . In particular a structure with bounded width admits a ternary WNU polymorphism.*

Because  $\text{Pol}(L_m)$  is idempotent (Lemma 2.2), Theorem 2.4 applies to  $L_m$  directly.

3. REDUCTION TO THE CUBE  $\{0, 1\}^3$ 

**Lemma 3.1** (Order-convex restriction). *Let  $m \geq 3$  and set  $T = \{0, 1\}^3 \times \{0\}^{m-3} = \{x \in \{0, 1\}^m : x_4 = \dots = x_m = 0\}$ . If  $f \in \text{Pol}(L_m)$  is ternary, then  $f(T \times T \times T) \subseteq T$  and the restriction  $f|_T$  is a ternary polymorphism of  $L_3$  under the poset isomorphism  $T \cong \{0, 1\}^3$ .*

*Proof.*  $T = [0^m, 1^3 0^{m-3}]$  is an interval, hence order-convex, so by Lemma 2.2  $f$  preserves it and  $f|_T: T^3 \rightarrow T$  is defined. The identification  $T \cong \{0, 1\}^3$  (drop the trailing zeros) is a poset isomorphism, so  $f|_T$  is monotone and idempotent. For a WNU  $f$  the restriction inherits the WNU identities on  $T$ . Finally  $f|_T$  is order-convex-preserving on  $T$ : if  $C \subseteq T$  is order-convex in  $T$ , then  $C$  is order-convex in  $\{0, 1\}^m$ —for  $a \leq c \leq b$  with  $a, b \in C \subseteq T$  and  $c \in \{0, 1\}^m$ , the order-convexity of  $T$  gives  $c \in T$ , and then order-convexity of  $C$  in  $T$  gives  $c \in C$ —so  $f$  preserves  $C$ , whence so does  $f|_T$ . By Lemma 2.2 (in dimension 3),  $f|_T \in \text{Pol}(L_3)$ .  $\square$

## 4. THE THEOREM

**Proposition 4.1** (Base case,  $m = 3$ ). *The clone  $\text{Pol}(L_3)$  contains no ternary weak near-unanimity operation.*

*Proof.* Suppose  $f \in \text{Pol}(L_3)$  is a ternary WNU. Write elements of  $\{0, 1\}^3$  as bit strings and  $m(x, y) := f(x, x, y) = f(x, y, x) = f(y, x, x)$  (the WNU identities). Two facts fix the values we need.

*Order-convex hulls.* By Lemma 2.2,  $f(a, b, c) \in \text{conv}\{a, b, c\}$ . For an antichain pair  $\{x, y\}$ ,  $\text{conv}\{x, y\} = \{x, y\}$  (a value  $\geq$  one of  $x, y$  and  $\leq$  one of them, with  $x, y$  incomparable, must be  $x$  or  $y$ ); so  $m(x, y) \in \{x, y\}$ . For the three coatoms 110, 101, 011 (a 3-antichain)  $\text{conv}\{110, 101, 011\} = \{110, 101, 011\}$  likewise, so  $f(110, 101, 011)$  is one of them. Monotonicity is  $a \leq a', b \leq b', c \leq c' \Rightarrow f(a, b, c) \leq f(a', b', c')$  (componentwise).

*The variables.* Consider the ten values

$$\begin{aligned} A &= m(010, 101) \in \{010, 101\}, & B &= m(100, 011) \in \{011, 100\}, \\ C &= m(101, 010) \in \{010, 101\}, & D &= m(101, 011) \in \{011, 101\}, \\ E &= m(101, 110) \in \{101, 110\}, & F &= m(001, 110) \in \{001, 110\}, \\ G &= f(110, 101, 011) \in \{011, 101, 110\}, & H &= m(110, 101) \in \{101, 110\}, \\ I &= m(110, 001) \in \{001, 110\}, & J &= m(110, 011) \in \{011, 110\}, \end{aligned}$$

the domains being the order-convex hulls just described. Reading off the componentwise order of the defining argument triples gives the monotonicity constraints

$$A \leq G, A \leq H, B \leq D, B \leq G, B \leq J, C \leq D, C \leq E, F \leq E, F \leq G, I \leq H, I \leq J.$$

(For instance  $F = f(110, 001, 001) \leq f(110, 101, 101) = E$  since  $001 \leq 101$  coordinatewise; and  $G = f(110, 101, 011)$  dominates  $A = f(010, 101, 010)$  since  $010 \leq 110, 101 \leq 101, 010 \leq 011$ .)

*Case  $B = 011$ .* Then  $D = 011$  ( $011 \not\leq 101$ ), so from  $C \leq D$ ,  $C = 010$ ; from  $C \leq E$ ,  $E = 110$  ( $010 \not\leq 101$ ); from  $F \leq E$ ,  $F = 110$  ( $001 \not\leq 110$ , its third bit); from  $B \leq G$ ,  $G = 011$  ( $011 \not\leq 101, 110$ ). But  $F \leq G$  now reads  $110 \leq 011$ , false.

*Case  $B = 100$ .* Then  $D = 101$ ; from  $C \leq D$ ,  $C = 101$  ( $010 \not\leq 101$ ); from  $C \leq E$ ,  $E = 101$  ( $101 \not\leq 110$ ); from  $F \leq E$ ,  $F = 001$  ( $110 \not\leq 101$ ); from  $F \leq G$ ,  $G = 101$  ( $001 \not\leq 110$ ); from  $A \leq G$ ,  $A = 101$ ; from  $A \leq H$ ,  $H = 101$ ; from  $I \leq H$ ,  $I = 001$  ( $110 \not\leq 101$ ); and from  $B \leq J$ ,  $J = 110$ . But  $I \leq J$  now reads  $001 \leq 110$ , false in the third bit.

Both cases are impossible, so no such  $f$  exists. (The twelve cells above are a minimal unsatisfiable core located by machine, `dedekind.o33_core.py`; the deduction is by hand. As a sanity check the same search on  $\{0, 1\}^2$  returns the median, and the full  $\{0, 1\}^3$  enumeration agrees: `dedekind.o30_wnu3.py`.) The obstruction is visible already in the median, the symmetric majority candidate, which fails because  $\text{med}(100, 010, 001) = 000$  escapes the hull of the three atoms.  $\square$

**Theorem 4.2.** *For every  $m \geq 3$ , the clone  $\text{Pol}(L_m)$  contains no ternary weak near-unanimity operation. Consequently  $L_m$  does not have bounded width.*

*Proof.* Suppose  $f \in \text{Pol}(L_m)$  were a ternary WNU,  $m \geq 3$ . By Lemma 3.1 its restriction  $f|_T$  is a ternary WNU in  $\text{Pol}(L_3)$ , contradicting Proposition 4.1. Hence no such  $f$  exists. Since  $\text{Pol}(L_m)$  is idempotent, the absence of a ternary WNU polymorphism denies the necessary condition of Theorem 2.4, so  $L_m$  does not have bounded width.  $\square$

The argument is elementary throughout: Proposition 4.1 is a finite hand computation on  $\{0, 1\}^3$  (machine only locating the twelve-cell core), and the passage to all  $m \geq 3$  is Lemma 3.1. An independent exhaustive search on  $\{0, 1\}^4$  returns the same non-existence, consistent with the theorem.

## 5. THE COMPLEXITY DICHOTOMY

Unbounded width says that no amount of *local consistency* decides  $\text{CSP}(L_m)$ ; it leaves open whether the problem is tractable by some other means (a Maltsev or few-subpowers algebra can be tractable without bounded width). The Bulatov–Zhuk dichotomy settles this: for a finite *idempotent* constraint language  $L$ ,  $\text{CSP}(L)$  is in P if  $\text{Pol}(L)$  is a Taylor clone, and NP-complete otherwise [5, 6]. Taylorness has a single-identity witness.

**Theorem 5.1** ([7, 8]). *A finite idempotent clone is Taylor if and only if it contains a 4-ary Siggers operation  $s$  satisfying*

$$s(a, r, e, a) = s(r, a, r, e) \quad \text{for all } a, r, e.$$

Since  $\text{Pol}(L_m)$  is idempotent,  $\text{CSP}(L_m)$  is in P iff  $\text{Pol}(L_m)$  contains a Siggers operation. A Siggers operation of  $\text{Pol}(L_m)$  is, by Lemma 2.2, monotone, idempotent and order-convex-preserving; and Lemma 3.1 applies verbatim to it (a 4-ary polymorphism preserves the order-convex set  $T = \{0, 1\}^3 \times \{0\}^{m-3}$  and restricts to one on  $\{0, 1\}^3$ , and the Siggers identity is inherited). So the existence of a Siggers operation on  $\{0, 1\}^m$  reduces, as before, to the single cube  $\{0, 1\}^3$ .

**Proposition 5.2.**  *$\text{Pol}(L_3)$  contains no Siggers operation.*

*Proof.* This is a finite satisfiability question—the monotone, idempotent, order-convex-preserving operations  $(\{0, 1\}^3)^4 \rightarrow \{0, 1\}^3$  satisfying the Siggers identity form a constraint problem in  $8^4$  cells with the values restricted to order-convex hulls, the diagonal fixed by idempotence, and cells identified by the Siggers identity. Encoded as a Boolean formula (exactly-one per cell class; the monotone order-network as binary clauses) it is UNSATISFIABLE (`dedekind_o34.sigsat.py`). The same encoding on  $\{0, 1\}^2$  is satisfiable—returning a Siggers operation whose defining identities are then independently rechecked—which validates it.  $\square$

**Theorem 5.3.** *For  $m \leq 2$  the problem  $\text{CSP}(L_m)$  is in P; for every  $m \geq 3$  it is NP-complete.*

*Proof.* For  $m \leq 2$  the median is a majority polymorphism of  $L_m$  (it preserves the order and every order-convex set, since on  $\{0, 1\}^{\leq 2}$  the median of any triple lies in the order-convex hull of its arguments), so  $L_m$  even has bounded width and  $\text{CSP}(L_m) \in \text{P}$ . For  $m \geq 3$ , a Siggers operation of  $\text{Pol}(L_m)$  would restrict to one of  $\text{Pol}(L_3)$  by the paragraph above, contradicting Proposition 5.2; so  $\text{Pol}(L_m)$  has no Siggers operation, is not Taylor by Theorem 5.1, and  $\text{CSP}(L_m)$  is NP-complete by [5, 6].  $\square$

The classification is summarised in Table 1. Three registers must be kept apart: whether local consistency decides the problem (bounded width); the complexity of the *uniform* problem, which has all order-convex relations at once; and the complexity of realizability along a *single* generator, which has only that generator’s fibres. The first two jump at  $m = 3$ ; the third is always polynomial.

The obstruction is thus sharper than a failure of local consistency: for  $m \geq 3$  the order-convex language is NP-hard. In the cubical setting this concerns the *uniform* realizability problem, in

|                                    | $m \leq 2$ | $m \geq 3$             |
|------------------------------------|------------|------------------------|
| median a majority polymorphism     | yes        | no                     |
| ternary WNU / bounded width        | yes        | no (Thm. 4.2)          |
| 4-ary Siggers / Taylor             | yes        | no (Prop. 5.2)         |
| CSP( $L_m$ ), uniform              | P          | NP-complete (Thm. 5.3) |
| realizability, one fixed generator | P          | P <sup>†</sup>         |

TABLE 1. The order-convex language  $L_m$  on  $\{0, 1\}^m$ : bounded width and the uniform complexity dichotomy jump between  $m = 2$  and  $m = 3$ , while realizability along any one fixed generator stays polynomial. Median-closure of the fibres is the exact boundary of the transition. <sup>†</sup>Polynomial in every computed instance; transparent for  $n \leq 3$ , where the structure retracts onto its image, a distributive lattice (Cor. 6.1).

which the generator and cell both vary: the variable domains are the fibres  $z^{-1}(a(v))$ , and as  $z$  ranges over all monotone maps these fibres range over all order-convex subsets of  $\{0, 1\}^m$  (every order-convex set is a fibre of some monotone map), so the uniform problem has exactly the language  $L_m$  and is NP-complete for  $m \geq 3$ . A *single* generator  $z$ , by contrast, presents only the sublanguage  $\{\leq\} \cup \{\text{fibres of } z\}$ , and this is tractable in every case we computed. When  $n \leq 3$  the image of a monotone map is always a distributive lattice and the structure  $(\{0, 1\}^m; \leq, \text{fibres of } z)$  retracts onto it (a monotone map collapsing each fibre to a representative), so it inherits the median and has strict width two; for  $n \geq 4$  the image need not be a lattice, yet the polymorphism clone still carries a Siggers operation and the problem stays in P throughout our tests. So the NP-hardness belongs to the uniform problem, where all order-convex relations are present at once, and not to any fixed generator—for which the unconditional consequence is the coskeletal statement below.

## 6. CONSEQUENCES FOR CUBICAL ATOMS

Combining Theorem 4.2 with the strict-width-two theorem of [1] isolates median-closure as an exact threshold.

**Corollary 6.1.** *Median-closure of the fibres is the exact boundary between bounded and unbounded width for the monotone-selection problems of the cube category:*

- (1) *if a generator  $z: \{0, 1\}^m \rightarrow \{0, 1\}^n$  has all fibres median-closed, its realizability problem has strict width two [1], and its atom is boundedly coskeletal;*
- (2) *if some fibre is order-convex but not median-closed, the median is no longer a polymorphism, and the uniform realizability problem—over all generators and cells—has the full order-convex language  $L_{m'}$  ( $m' \geq 3$ ), which has unbounded width and is NP-hard (Theorems 4.2, 5.3); the atom of such a generator is unboundedly coskeletal. (Realizability along one fixed generator, by contrast, was polynomial in every case we computed—for  $n \leq 3$  transparently, its structure retracting onto its image, which is then a distributive lattice; so the NP-hardness is a property of the uniform problem, where all order-convex relations are available at once, not of any single generator.)*

*In particular, the finiteness of the atom family of  $\square^n$  obtained from strict width two is confined to the median-closed class in every dimension.*

This is the negative half of the finiteness question for the Dedekind comparison: the bounded-width lever cannot reach non-median-closed generators, so a finiteness argument for the general atom order must be sought elsewhere, and the shape-by-shape assembly is correspondingly not the endgame—consistent with the reduction of the comparison to a single cubical approximation statement in [1].

## 7. DATA AVAILABILITY

The twelve-cell minimal core of Proposition 4.1 was located by `dedekind_o33_core.py`; the deduction from it is by hand. The corroborating searches are `dedekind_o30_wnu3.py` (the exhaustive weak near-unanimity search on  $\{0, 1\}^3$  and  $\{0, 1\}^4$ , with the  $\{0, 1\}^2$  validation returning the median). The Siggers non-existence of Proposition 5.2 is the UNSAT verdict of the SAT encoding `dedekind_o34_sigsat.py` (validated by the satisfiable  $\{0, 1\}^2$  instance). All are in the version-pinned artifact accompanying [1].

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